

- 17** Chloroethene, $\text{CH}_2=\text{CHCl}$, is unreactive to nucleophiles whereas 3-chloropropene, $\text{CH}_2=\text{CHCH}_2\text{Cl}$, is very reactive.

Which statement best explains the lack of reactivity of $\text{CH}_2=\text{CHCl}$ to nucleophiles?

- A** All substituted alkenes undergo only electrophilic addition.
- B** The $\text{C}=\text{C}$ double bond and the Cl atom are both electron withdrawing which stabilises $\text{CH}_2=\text{CHCl}$.
- C** Electrons in a p orbital of the chlorine atom overlap with the π bond.
- D** The presence of the π bond prevents free rotation of the $\text{C}-\text{Cl}$ bond thus decreasing the reactivity.